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What Differentiates Elite, Sub-Elite, and Amateur Young Soccer Players? A Comparative Analysis of Match-Play Locomotor Demands

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Abstract

This observational cross-sectional study compared match-play locomotor demands among youth soccer players competing at three competitive levels (elite, sub-elite, and amateur) within the same club. Fifty-five male U13–U14 soccer players participated in the study (elite, n = 19; sub-elite, n = 17; amateur, n = 19). Ten official matches were analyzed during the 2024–2025 season (four elite, three sub-elite, and three amateur matches), with external load monitored using 10-Hz GPS devices. Locomotor variables were compared among competitive levels using multivariate and univariate analyses of covariance (MANOVA/ANCOVA), with Peak Height Velocity (PHV) as a continuous covariate, followed by post-hoc comparisons. No significant between-group differences were observed for total distance covered or distances performed at low-to-moderate running speeds. In contrast, differences became more apparent at higher running velocities. Elite players covered greater absolute distances at high-speed thresholds, particularly at velocities ≥24 km·h^{-1}, and achieved higher maximal speeds and absolute sprint distances than sub-elite and amateur players. Differences were also observed in acceleration–deceleration profiles, with sub-elite players performing a greater number of total acceleration and deceleration events, whereas elite players achieved higher maximal acceleration and deceleration values and covered greater distances within high-intensity acceleration zones. Conversely, while impact-related measures showed limited ability to differentiate among competitive levels, multi-planar aggregated load metrics—specifically Player Load and its directional vector components—robustly discriminated between playing tiers once adjusted for biological maturation. Overall, the findings indicate that high-intensity locomotor variables, particularly absolute high-speed running and maximal speed, differed more consistently among the three competitive groups than total movement volume or impact-related measures. Crucially, these findings reflect an independent effect of competitive standard rather than mere differences in developmental status. Locomotor demands may therefore provide useful complementary information for profiling youth soccer players, offering a

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robust framework for competitive differentiation that can be further validated in future maturity-matched and longitudinal studies.

Keywords: external load; locomotor demands; GPS devices; soccer; talents

1. Introduction

Performance in soccer is inherently multifactorial, depending on technical and tactical abilities as well as the physical and locomotor demands imposed throughout match play [1,2]. External load provides an objective measure of the work performed by athletes during training and competition [3]. Common external-load indicators, including speed, acceleration, deceleration, power output, and time–motion variables [3], provide relevant information about players' physical responses under real match conditions. For instance, total distance covered and high-intensity running during official matches have been associated with aerobic fitness [4,5]. Consequently, monitoring match demands has become an important component of performance assessment across competitive levels [3]. Locomotor data may also provide complementary information within talent identification (TID) and player-development processes by helping characterize the physical demands associated with different competitive environments [6].

Physical fitness is an important determinant of a player's ability to cope with match locomotor demands. Previous research has associated body composition [7,8], endurance [9] cardiorespiratory fitness [10,11] and change-of-direction (COD) ability [12–14] with soccer performance. Mohr et al. [15], for example, reported that top-class players performed more high-intensity running during matches and achieved better Yo-Yo test performance than moderate-level professionals. Similarly, Oliva et al. [16] observed fewer sprint actions in youth than in professional players. Collectively, these findings suggest that physical fitness and match locomotor performance are closely related, supporting the potential value of combining fitness and locomotor information when profiling developing players [17,18].

However, match running performance is influenced by several individual and contextual factors. Age [19–22], playing position [20,22–24], competitive level [6,25,26] or team formation [27] have all been identified as relevant determinants of variability in locomotor demands. For example, Filipa et al. [28] reported that older players are faster, possess higher maximal sprint speed and anaerobic speed reserve, and demonstrate superior COD performance compared to younger cohorts. Similarly, comparisons between U14 and U15 players have shown greater high-speed running performance in older age groups [19]. In a longitudinal study from U9 to U18 players, Saward et al. [20] also demonstrated that playing position influenced match running performance and its age-related development, with central defenders generally covering less distance and wide players more, reflecting the different physical and tactical demands associated with positional roles.

Competitive level and opposition quality may further shape the locomotor demands experienced during match play. Kádár et al. [22] observed that U16, U19, and U21 players competing against stronger opponents tended to perform more high-intensity running, although differences were not statistically significant. Similarly, lower-level U17 teams have been reported to cover less sprint distance than top- and middle-level teams [24], while total distance, running at 4.4–5.5 $m \cdot s^{-1}$, and acceleration-related variables have been associated with team success [29]. However, contrasting evidence exists in professional soccer, Modric et al. [30] found that players' physical performance was similar regardless of whether they competed against high- or low-quality teams, and Di Salvo et al. [31] reported greater sprint distances in teams ranked outside the Premier League top five. Thus,

the relationship between competitive level and locomotor performance appears to be complex and may depend on the interaction of individual, developmental, competitive, and contextual factors.

Competitive level may also be associated with differences in tactical performance. More experienced players tend to demonstrate more coordinated tactical behaviors and more effective perceptual responses to changing match situations [32]. Such expertise may influence locomotor performance because tactical decisions determine when, where, and under which circumstances high-intensity actions are performed. Consequently, similar overall running volumes may represent different performance responses depending on their tactical context. Decision-making, positioning, and game understanding have also been identified as relevant components of soccer performance and player development [33], while cognitive, psychological, and perceptual characteristics are increasingly considered within multidimensional approaches to TID [17,34]. The Nomination Scale for Identifying Football Talent (NSFIT), for example, provides a framework for assessing several of these dimensions from the perspectives of players, parents, and coaches [34].

The combined consideration of physical, locomotor, and contextual dimensions may therefore provide a more comprehensive understanding of youth soccer performance. Monitoring data can support coaches and practitioners in evaluating match demands, training load, fatigue, and recovery and can inform decisions regarding periodization and match preparation [35,36]. However, despite the recognized influence of competitive level and tactical factors on match performance, evidence directly comparing match-derived locomotor profiles across competitive levels under relatively standardized club conditions remains limited. Furthermore, locomotor actions should not be interpreted independently of the match context in which they occur, as similar physical outputs may reflect different performance demands [37,38]. Although tactical behavior represents an important component of this broader framework, it was not directly assessed in this study and was therefore considered a contextual factor rather than an outcome variable. Accordingly, the primary aim of this study was to compare match-play locomotor demands among youth soccer players competing at three different levels (amateur, sub-elite, and elite) within the same club. A secondary aim was to identify the locomotor variables that most clearly differentiate these competitive levels and may therefore provide complementary information for player profiling and development.

2. Materials and Methods

2.1. Participants

This observational, cross-sectional study included a sample of 55 young male soccer players from the same club with players from 3 different teams of the same age group (U13-14) that compete in three different categories: amateur, low level (19 players), sub-elite, 2nd highest level (17 players) and elite, highest level (19 players). Table 1 shows descriptive data on the age and anthropometry characteristics of players in each team. It also includes data from physical fitness tests to assess performance differences between the teams.

The eligibility criteria for participation included: (i) possessing an official license issued by the Royal Spanish Football Federation (RFEF), ensuring official recognition and adherence to regulatory standards; (ii) demonstrating the absence of any physical conditions or injuries that could impair performance or pose health risks during testing, as confirmed through medical screening; and (iii) obtaining written informed consent from legal guardians, in accordance with ethical research protocols and to ensure understanding of the study procedures and potential risks involved. All players actively participated throughout the entire season, and if data loss occurred in any of the variables (absence during one match or training for injury or not called up), these data were not included in

the analysis, ensuring a complete a representative dataset. See Table 1, for more information.

For each locomotor variable, values obtained across eligible match observations were averaged for each player. Thus, each player contributed a single value per variable to the statistical analyses, and the player, rather than the individual match observation, constituted the unit of analysis.

Table 1. Descriptive data of the group (anthropometrical measures and physical fitness test assessment [CMJ, SJ, and Handgrip test]) in amateur, sub-elite and elite youth soccer players.

Variable	Amateur (Mean ± SD) *n* = 19	Sub-elite (Mean ± SD) *n* = 17	Elite (Mean ± SD) *n* = 19
Age (years)	12.84 ± 0.56	13.58 ± 0.25	13.89 ± 0.19
Age PHV (years)	14.14 ± 0.42	14.18 ± 0.70	13.63 ± 0.61
Maturity offset	−1.30 ± 0.57	−0.56 ± 0.77	0.27 ± 0.55
Anthropometrical measures			
Height (cm)	154.82 ± 8.93	159.50 ± 9.44	169.34 ± 4.32
Weight (kg)	40.82 ± 17.72	47.01 ± 8.22	61.20 ± 6.72
Physical Fitness			
CMJ (cm)	21.98 ± 5.65	27.24 ± 5.12	32.50 ± 2.69
SJ (cm)	20.52 ± 5.46	26.60 ± 4.68	30.66 ± 2.88
Handgrip right (kg)	23.35 ± 5.51	26.44 ± 7.76	34.46 ± 4.67
Handgrip left (kg)	21.38 ± 6.17	25.68 ± 7.09	31.67 ± 6.61

2.2. Procedure

Training and match evaluations took place during the data collection period. Athletes performed a pre-match warm-up, including running and mobility, to prepare them for the following test: (i) the countermovement jump (CMJ) test and (ii) the handgrip test. These tests were assessed in the dressing room just after anthropometry. Weight and height (stand and seated height for obtaining PHV and Maturity Offset) were evaluated with subjects dressed only in their training pants to ensure accuracy and consistency.

Two weeks (four days) were needed to evaluate the amateur and sub-elite teams (which train on the same field at the same time), and another week (two days) was needed for the elite team. When players were not being evaluated, they continued with their regular coach-led activities. Once the tests were completed, they returned to the team training. A similar structure was followed across the evaluation of all three teams. Before testing, players were given verbal information and examples from the research team about the protocol of each test. A familiarization attempt was performed by participants for each test.

Match evaluation took place during the 2024–2025 season. Ten matches were analyzed (3 for amateur and sub-elite teams and 4 for the elite team) and locomotor demands were obtained. Each match lasted 80 min (two 40′ halves). Only players that completed more than 50 min were included in the analysis. All teams played with a free-change policy and with the same 4-3-3 formation. Goalkeepers were excluded from the locomotor assessment.

External-load measures were assessed with GPS devices (WIMU PROTM, RealTrack Systems, Almería, Spain) operating at a sampling frequency of 10 Hz (Muñoz-López et al., 2017 [39]). Microsensor units were harnessed within a tight-fitting vest that was worn by soccer players during the study. The microsensor devices were activated 15 min before the start of each testing match, in accordance with the manufacturer's recommendations. Data were downloaded after the intervention protocol to a computer and analyzed using a customized software package (WIMU SPRO, Almería, Spain, SPRO version 1.0.0

compilation: 969). In addition, the GPS devices were coupled with a 100 Hz triaxial accelerometer that allowed for the estimation of body impact data provided in "g" force and the calculation of load measures. The equipment presented good levels of validity and reliability in linear, circular and zig-zag courses of Bastida-Castillo et al., 2019 [40]. Data were collected under what were considered good weather and satellite conditions for GPS use (number of satellites = 10.1 ± 0.2).

2.3. Locomotor Demands

Variables were divided into distance, acceleration–decelerations, sprint-related, impact and load variables. The name, explanation and unit of each variable are presented in Table 2.

Table 2. Analyzed locomotor demands variables during official matches.

Variable	Explanation	Unit
Distance		
Total Distance	Total distance covered	m
Explosive Distance	Total distance covered with acceleration > 1.12 $m \cdot s^{-2}$	m
DZ1	Total distance covered at 0–6 $km \cdot h^{-1}$	m
DZ2	Total distance covered at 6–12 $km \cdot h^{-1}$	m
DZ3	Total distance covered at 12–18 $km \cdot h^{-1}$	m
DZ4	Total distance covered at 18–21 $km \cdot h^{-1}$	m
DZ5	Total distance covered at 21–24 $km \cdot h^{-1}$	m
DZ6	Total distance covered at 24–27 $km \cdot h^{-1}$	m
DZ7	Total distance covered at 27–30 $km \cdot h^{-1}$	m
DZ8	Total distance covered at 30–50 $km \cdot h^{-1}$	m
Accelerations and Decelerations		
No. of Acc	Total number of accelerations (>0 $m \cdot s^{-2}$)	N
No. of Dcc	Total number of decelerations (<0 $m \cdot s^{-2}$)	N
Maximum Acceleration	Maximum acceleration reached	$m \cdot s^{-2}$
Maximum Deceleration	Maximum deceleration reached	$m \cdot s^{-2}$
Average Acceleration	Mean acceleration intensity	$m \cdot s^{-2}$
Average Deceleration	Mean deceleration intensity	$m \cdot s^{-2}$
High-Intensity Accelerations	Total distance covered during high-intensity accelerations (>3 $m \cdot s^{-2}$)	m
High-Intensity Decelerations	Total distance covered during high-intensity decelerations (<–3 $m \cdot s^{-2}$)	m
Sprints		
Absolute Sprints	Total distance covered above the absolute sprint threshold (24 $km \cdot h^{-1}$)	m
Relative Sprints	Total distance covered above the relative sprint threshold (95% of individual max speed)	m
Absolute High-Speed Running	Total distance covered above the absolute HSR threshold (21 $km \cdot h^{-1}$)	m
Relative High-Speed Running	Total distance covered above the relative HSR threshold (75.5% of individual max speed)	m
Maximal Speed	Maximum speed reached	$m \cdot s^{-1}$
Average Speed	Mean speed	$m \cdot s^{-1}$
Impacts		

Total Impacts	Total number of acceleration events exceeding 2 g	N
High-Impact Count	Total number of high-intensity impacts (>8 g) in the horizontal plane	Number
Load		
Player Load	Cumulative measure of external load based on triaxial accelerations recorded by the WIMU device	AU
Horizontal Player Load	Player Load index in the horizontal axis; accumulation of movement in horizontal plane	AU
Vertical Player Load	Player Load index in the vertical axis; accumulation of movement in vertical plane	AU
Anteroposterior Player Load	Player Load index in the anteroposterior axis; accumulation of movement in this plane	AU
Mediolateral Player Load	Player Load index in the mediolateral axis; accumulation of movement in this plane	AU
Metabolic Power	Metabolic power output	$W \cdot kg^{-1}$
Average Metabolic Power	Mean metabolic power	$W \cdot kg^{-1}$
High Metabolic Load Distance	Distance covered at high metabolic power (>25.5 $W \cdot kg^{-1}$)	m
Energy Expenditure	Total energy expenditure associated with activities, including accelerations and decelerations	kcal
Maximum Acceleration	Maximum instantaneous acceleration recorded	$m \cdot s^{-2}$
Max Acc Speed	Speed at the point of maximum acceleration	$m \cdot s^{-1}$

2.4. Statistical Analysis

Statistical analyses were performed using JASP (version 0.19.3; JASP Team, 2025 [41]). Descriptive statistics, including means, standard deviations, medians, interquartile ranges, minimums, and maximums, were calculated for all dependent variables.

To examine differences in locomotor demands among competitive levels (amateur, sub-elite, and elite), multivariate analyses of variance (MANOVAs) were conducted for conceptually related sets of dependent variables, with the competitive level included as the between-subject factor. Multivariate effects were evaluated using Pillai's trace. Homogeneity of variance was assessed using Levene's test.

To control for the potentially confounding effect of biological maturation on locomotor performance, Peak Height Velocity (PHV) was included as a continuous covariate. Consequently, when a significant multivariate effect was identified, follow-up univariate analyses of covariance (ANCOVAs) were conducted to examine the specific variables contributing to the overall group effect, while statistically isolating the influence of PHV. Omega-squared (ω^2) was calculated as an estimate of effect size for the univariate analyses and interpreted as small (0.010–0.059), medium (0.060–0.149), or large (≥0.150) (Cohen, 1988 [42]). When significant between-group differences were identified after adjusting for the covariate, post-hoc pairwise comparisons were performed using Tukey's Honest Significant Difference (HSD) test on the estimated marginal means. The magnitude of pairwise differences was also quantified using Cohen's *d* and interpreted as trivial (≤0.20), small (>0.20–0.60), moderate (>0.60–1.20), large (>1.20–2.00), or very large (>2.00–4.00) (Hopkins, 2002 [43]). Statistical significance was set at $p < 0.05$.

3. Results

3.1. Locomotor Demands Comparison Between Teams

Distance

A multivariate analysis of variance (MANOVA) was initially conducted to evaluate differences in overall distance variables covered between elite, sub-elite, and amateur teams. The MANOVA results indicated a significant main effect of team level on the set of distance variables (Pillai's Trace = 0.766, F (18, 88) = 3.036, $p < 0.001$).

Following this omnibus effect, separate univariate analyses of covariance (ANCOVAs) were conducted for each distance metric, using Peak Height Velocity (PHV) as a continuous covariate to control for biological maturation. As shown in Table 3, the individual ANCOVAs revealed no statistically significant differences among the competitive levels for TD ($p = 0.71$), ED ($p = 0.521$), 0–6 km/h ($p = 0.106$), 6–12 km/h ($p = 0.193$), and 12–18 km/h ($p = 0.178$). Conversely, significant main effects of competitive level were identified for all high-speed locomotor categories starting from 18 km/h. Specifically, the individual ANCOVAs revealed medium main effects for the 18–21 km/h range ($p = 0.013$, $\omega^2 = 0.125$) and the 21–24 km/h range ($p = 0.009$, $\omega^2 = 0.132$). Furthermore, large main effects emerged for both the 24–27 km/h range ($p < 0.001$, $\omega^2 = 0.216$) and the 27–30 km/h range ($p < 0.001$, $\omega^2 = 0.213$). The covariate PHV did not show a significant association with any of these high-speed distance metrics ($p \geq 0.05$).

Table 3. Descriptive statistics (mean +/− sd) for distance metrics for amateur, sub-elite and elite teams, with univariate ANCOVA *p*-values, and effect size ω^2.

Variable	Amateur	Sub-Elite	Elite	*p*-Value	ω^2
TD (m)	6564 ± 1355	6624 ± 1784	7114 ± 1158	0.71	-
ED (m)	624.2 ± 216.8	672.6 ± 247.7	694 ± 151.5	0.521	-
[0–6] km/h (m)	3157 ± 673.7	2873 ± 607.1	3479 ± 631.4	0.106	-
[6–12] km/h (m)	2214 ± 521.2	2312 ± 789.6	2084 ± 425.4	0.193	-
[12–18] km/h (m)	1010 ± 318.5	1161 ± 437.6	1192 ± 303.3	0.178	-
[18–21] km/h (m)	131.8 ± 63.83	190.30 ± 113.3	213.8 ± 61.45	0.013 *	0.123
[21–24] km/h (m)	43.23 ^{a} ± 8.741	68.25 ^{ab} ± 22.41	93.66 ^{b} ± 29.57	0.009 **	0.132
[24–27] km/h (m)	7.387 ^{a} ± 1.057	17.06 ^{ab} ± 5.164	38.02 ^{c} ± 13.70	<0.001 ***	0.216
[27–30] km/h (m)	0.378 ^{a} ± 0.00	2.480 ^{ab} ± 0.269	11.81 ^{c} ± 4.42	<0.001 ***	0.213
[30–50] km/h (m)	-	0.098 ± 0.26	1.84 ± 4.42	-	-

Note: *p*-value refers to the *p*-value of the univariate ANCOVA test for the main effect of Team Level when controlling Biological Maturation (PHV). ω^2 represents the effect size; only reported for significant effects ($p < 0.05$). * $p < 0.05$. ** $p < 0.01$. *** $p < 0.001$. - value means not reported/calculated by JASP for non-significant results. Means that do not share a common superscript letter (a, b, c) are significantly different from one another according to Tukey's HSD post-hoc test (e.g., for [24–27] km/h, Elite is different from sub-elite and amateur; sub-elite is different from elite).

Post-hoc pairwise comparisons on the estimated marginal means tracked a clear upward trend, demonstrating that higher competitive levels covered greater distances as the speed thresholds increased (elite more than sub-elite, and sub-elite more than amateur (see letters in Table 3). In the 18–21 km/h range, the elite team presented moderate differences compared to the amateur team (d = 0.94, $p = 0.019$), while the sub-elite team showed a non-significant moderate difference compared to amateurs (d = 0.82, $p = 0.052$), and trivial differences were found between sub-elite and elite teams (d = 0.12, $p = 0.936$). In the 21–24 km/h range, moderate differences were found between amateur and elite teams (d = 1.06, $p = 0.007$). Conversely, non-significant moderate differences were

observed between amateur and sub-elite teams (d = 0.64, p = 0.159), and small differences were identified between sub-elite and elite cohorts (d = 0.42, p = 0.466). For the higher-speed thresholds, the distinction among competitive levels became significantly clearer. In the 24–27 km/h range, small non-significant differences were found between amateur and sub-elite teams (d = 0.55, p = 0.247). However, moderate differences were identified between sub-elite and elite teams (d = 0.86, p = 0.048), and large differences were revealed between amateur and elite players (d = 1.42, p < 0.001). A similar pattern occurred in the 27–30 km/h range, showing small non-significant differences between amateur and sub-elite teams (d = 0.31, p = 0.636), moderate differences between sub-elite and elite teams (d = 1.03, p = 0.015), and large differences between amateur and elite cohorts (d = 1.34, p < 0.001). Finally, sprinting distances within the 30–50 km/h range were exclusively registered by the elite team (1.84 ± 4.42 m). See Table 3 and Figure 1 for more information.

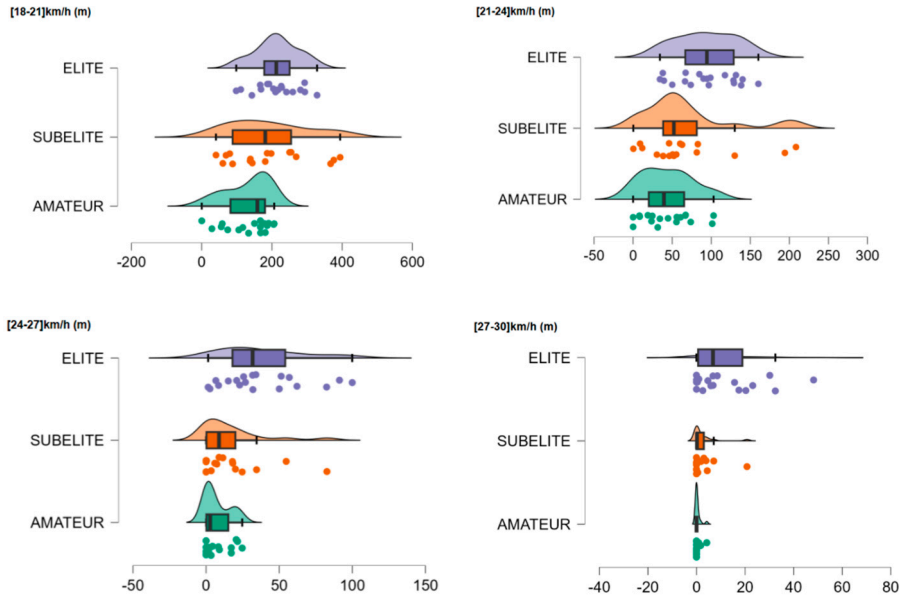


Figure 1. Raincloud plots showing speed-range distance (m) by team level. Each group includes raw data points, a density curve, and a boxplot displaying the median and interquartile range. Note: Most players (especially in the amateur and sub-elite levels) recorded 0 m in [27–30] km/h and [30–50] km/h (m) ranges, so raincloud plots are not suitable.

3.2. Accelerations–Decelerations

The MANOVA results indicated a significant main effect of team level on the set of acceleration and deceleration variables (Pillai's Trace = 1.04, F(16, 90) = 6.06, p < 0.001). As shown in Table 4, the individual ANCOVAs revealed no statistically significant main effects of competitive level on the average metrics, including Avg Acc (p = 0.403) and Avg Dcc (p = 0.358). Furthermore, after controlling for biological maturation, the main effect for High Dcc (m) did not reach statistical significance (p = 0.079, ω^2 = 0.059). In contrast, highly statistically significant main effects of competitive level persisted for the remaining metrics, revealing large effects for No. Acc (p < 0.001, ω^2 = 0.291), No. Dcc (p < 0.001, ω^2 = 0.28), Max Acc (p = 0.006, ω^2 = 0.155), Max Dec (p = 0.002, ω^2 = 0.196), and High Acc (m) (p < 0.001, ω^2 = 0.231). The covariate PHV did not show a significant association with any of the acceleration or deceleration metrics ($p \geq 0.060$).

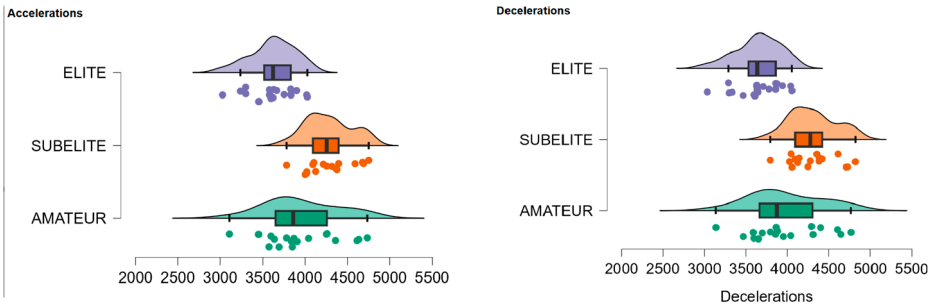
Table 4. Descriptive statistics (mean +/− sd) for acceleration/deceleration metrics for amateur, sub-elite and elite teams, with univariate ANCOVA *p*-values, and effect size ω^2.

Variable	Amateur	Sub-Elite	Elite	*p*-Value	ω^2
No. Acc	3955 ^{a} ± 442.80	4281 ^{b} ± 278.10	3619 ^{a} ± 266.90	<0.001 ***	0.291
No. Dcc	3975 ^{a} ± 445.10	4307 ^{b} ± 285.40	3647 ^{a} ± 272.10	<0.001 ***	0.28
Max Acc (m·s^{-2})	4.24 ^{a} ± 0.78	4.46 ^{a} ± 0.31	4.97 ^{b} ± 0.42	0.006 **	0.155
Max Dcc (m·s^{-2})	−5.11 ^{a} ± 0.71	−5.25 ^{a} ± 0.57	−5.96 ^{b} ± 0.63	<0.002 **	0.196
Avg Acc (m·s^{-2})	0.39 ± 0.09	0.34 ± 0.07	0.38 ± 0.07	0.403	-
Avg Dcc (m·s^{-2})	−0.40 ± 0.09	−0.35 ± 0.07	−0.39 ± 0.08	0.358	-
High Acc (m)	109.80 ^{a} ± 62.23	166.60 ^{ab} ± 106.82	233.80 ^{b} ± 69.35	<0.001 ***	0.231
High Dcc (m)	95.53 ± 53.31	96.13 ± 53.13	137.06 ± 45.51	0.079	-

Note: *p*-value refers to the *p*-value of the univariate ANCOVA test for the main effect of Team Level. ω^2 represents the effect size; only reported for significant effects ($p < 0.05$). ** $p < 0.01$. *** $p < 0.001$. - value means not reported/calculated by JASP for non-significant results. Means that do not share a common superscript letter (a, b) are significantly different from one another according to Tukey's HSD post-hoc test.

Post-hoc pairwise comparisons on the estimated marginal means revealed distinct patterns across competitive levels. For the volume metrics (No. Acc and No. Dcc), the sub-elite category exhibited the highest counts, showing large differences compared to the elite team (d = 1.75, $p < 0.001$ for No. Acc; d = 1.717, $p < 0.001$ for No. Dcc) and moderate differences compared to the amateur cohort (d = 0.996, $p = 0.014$ for No. Acc; d = 1.003, $p = 0.013$ for No. Dcc). No statistically significant differences in volume metrics were found between the amateur and elite teams ($p \geq 0.070$).

Regarding maximum performance metrics, the elite team demonstrated higher values, showing differences approaching a large effect for both Max Acc and Max Dec relative to the amateur team (d = 1.09, $p = 0.006$; d = 1.18, $p = 0.003$, respectively) and moderate differences compared to the sub-elite team (d = 0.87, $p = 0.046$; d = 1.09, $p = 0.01$, respectively). Trivial-to-small non-significant differences appeared between the amateur and sub-elite groups in these maximum metrics (d ≤ 0.22, $p \geq 0.797$). For distances covered at high acceleration (High Acc), the elite team registered large differences compared to the amateur team (d = 1.437, $p < 0.001$). Conversely, the sub-elite team showed a non-significant moderate difference compared to the amateur team (d = 0.801, $p = 0.058$) and small non-significant differences versus the elite group (d = 0.634, $p = 0.183$). See Table 4 and Figure 2 and for further details.



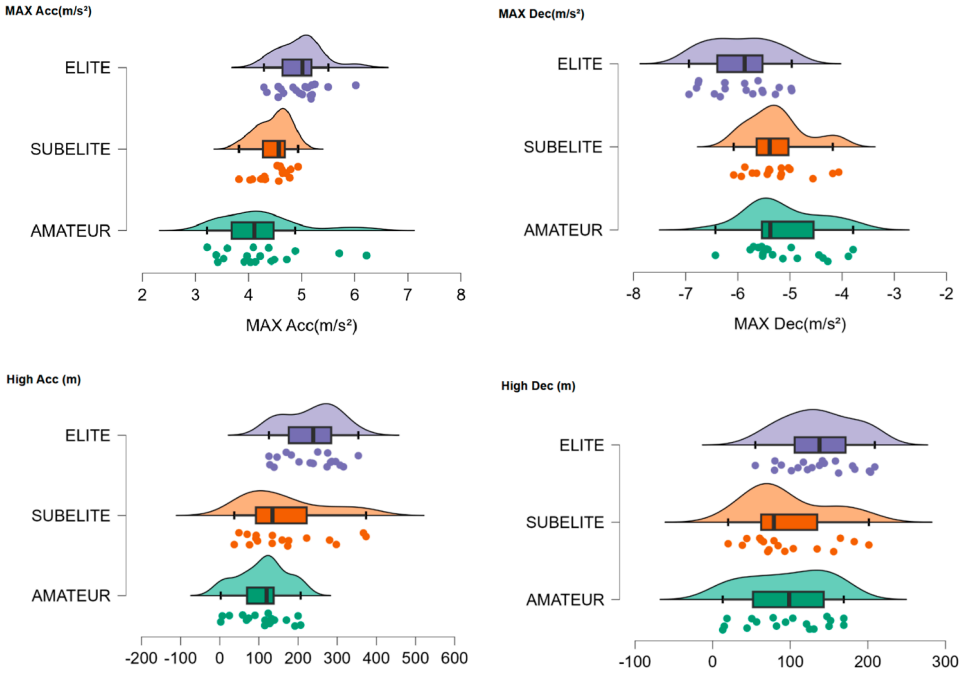


Figure 2. Raincloud plots showing acceleration and deceleration metrics by team level. Each group includes raw data points, a density curve, and a boxplot displaying the median and interquartile range.

3.3. Sprints

Sprint and speed analysis revealed a significant multivariate main effect of competitive level (Pillai's Trace = 0.49, F(12, 94) = 2.57, $p = 0.005$). As shown in Table 5, the individual ANCOVAs showed no statistically significant differences among competitive levels for the relative metrics, including Sprint Rel ($p = 0.18$), HSR Rel ($p = 0.089$), and AVG Speed ($p = 0.137$). In contrast, highly statistically significant main effects of competitive level were found for the absolute variables, revealing large effects for Sprint Abs ($p < 0.001$, $\omega^2 = 0.197$), HSR Abs ($p = 0.003$, $\omega^2 = 0.160$), and MAX Speed ($p < 0.001$, $\omega^2 = 0.270$). Notably, the covariate PHV demonstrated a significant association exclusively with the absolute volumetric metrics, including Sprint Abs ($p = 0.029$, $\omega^2 = 0.057$) and HSR Abs ($p = 0.032$, $\omega^2 = 0.057$), while showing no significant relationship with MAX Speed or any relative metric ($p \geq 0.175$).

Table 5. Descriptive statistics (mean +/− sd) for sprint metrics for amateur, sub-elite and elite teams, with univariate ANCOVA *p*-values, and effect size ω^2.

Variable	Amateur	Sub-Elite	Elite	*p*-Value	ω^2
Sprint Abs (m)	21.54 ^{a} ± 23.30	35.91 ^{a} ± 44.94	80.62 ^{b} ± 61.20	<0.001 ***	0.197
Sprint Rel (m)	51.17 ± 17.92	38.98 ± 20.20	42.52 ± 14.22	0.18	-
HSR Abs (m)	88.10 ^{a} ± 58.11	142.12 ^{ab} ± 123.51	205.24 ^{b} ± 94.28	0.003 **	0.16
HSR Rel (m)	344.80 ± 90.49	335.70 ± 143.50	290.90 ± 75.99	0.089	-
Max Speed (km/h)	24.05 ^{a} ± 2.37	25.79 ^{a} ± 2.28	28.23 ^{b} ± 3.04	<0.001 ***	0.27
Avg Speed (km/h)	5.27 ± 0.39	5.49 ± 0.44	5.52 ± 0.50	0.137	-

Note: *p*-value refers to the *p*-value of the univariate ANCOVA test for the main effect of Team Level. ω^2 represents the effect size; only reported for significant effects ($p < 0.05$). ** $p < 0.01$. *** $p < 0.001$. - value means not reported/calculated by JASP for non-significant results. Means that do not share a common superscript letter (a, b) are significantly different from one another according to Tukey's HSD post-hoc test.

Post-hoc pairwise comparisons highlighted distinct group variations. For Sprint Abs, the elite team covered significantly greater distances than both the amateur (d = 1.318, $p < 0.001$; large effect) and sub-elite teams (d = 0.913, $p = 0.034$; moderate effect), whereas no significant difference was observed between the amateur and sub-elite teams (d = 0.405, $p = 0.466$; small effect). Regarding HSR Abs, the elite team also recorded significantly higher values than the amateur players (d = 1.184, $p = 0.002$; moderate-to-large effect), while the elite–sub-elite (d = 0.509, $p = 0.332$) and amateur–sub-elite (d = 0.676, $p = 0.127$) comparisons did not reach statistical significance. For MAX Speed, the elite players achieved significantly higher velocities than both the amateur (d = 1.562, $p < 0.001$; large effect) and sub-elite groups (d = 0.914, $p = 0.034$; moderate effect). Although a moderate standardized difference was observed between the amateur and sub-elite teams (d = 0.648), this specific comparison was not statistically significant ($p = 0.149$). These findings are visually synthesized in Figure 3, with comprehensive pairwise metrics available in Table 5.

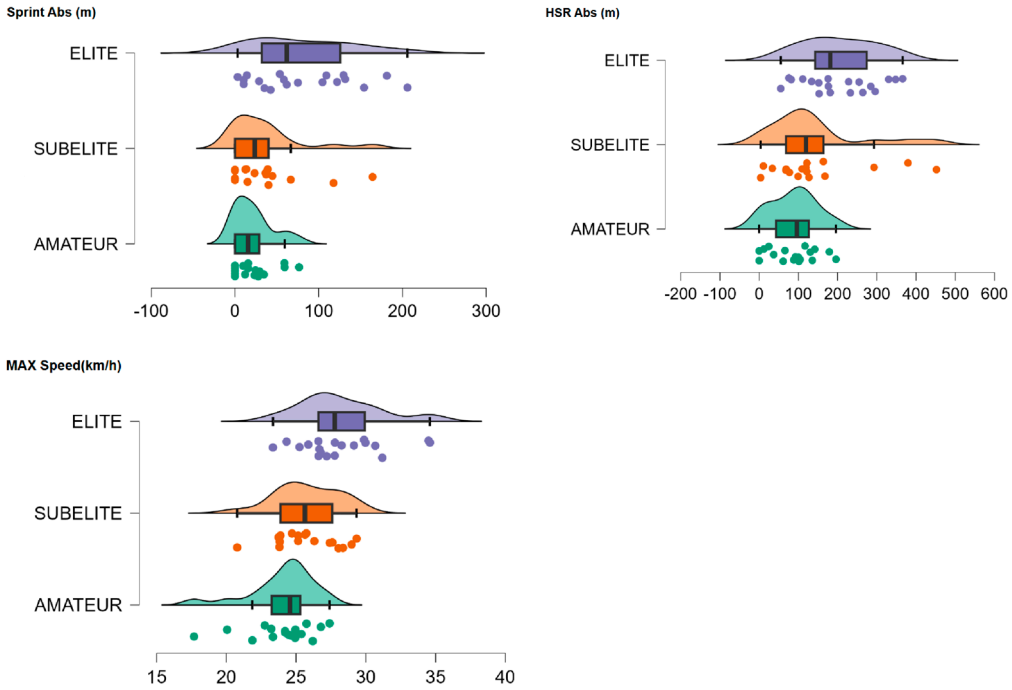


Figure 3. Raincloud plots showing sprint metrics by team level. Each group includes raw data points, a density curve, and a boxplot displaying the median and interquartile range.

3.4. Impacts and Load

Separate MANOVAs were conducted for the impact and load metrics. For the impact variables, the MANOVA did not indicate a significant main effect of competitive level (Pillai's Trace = 0.09, F(4, 102) = 1.21, $p = 0.31$). Consistent with the multivariate result, the subsequent univariate ANCOVAs showed no significant differences among groups for Total Impacts ($p = 0.21$) or High-Impact Count ($p = 0.256$). In contrast, the MANOVA for the load variables did indicate a significant effect (Pillai's Trace = 0.71, F (18, 88) = 2.69, $p = 0.001$). Following previous methodology, separate univariate analyses of covariance (ANCOVAs) confirmed significant main effects of competitive level across all Player Load variables. Specifically, medium main effects of team level were identified for total Player Load ($p = 0.012$, $\omega^2 = 0.115$), PL horizontal ($p = 0.015$, $\omega^2 = 0.108$), PL vertical ($p = 0.021$, $\omega^2 = 0.100$), PL anteroposterior ($p = 0.012$, $\omega^2 = 0.115$), and PL mediolateral ($p = 0.025$, $\omega^2 = 0.091$). Conversely, no statistically significant differences among competitive levels were

observed for Power met, Power met avg, HMLD or Energy Expenditure ($p > 0.073$). Unlike the previous distance variables, the covariate PHV demonstrated a significant association with most of these individual load metrics, including total Player Load ($p = 0.010$, $\omega^2 = 0.092$), PL horizontal ($p = 0.009$, $\omega^2 = 0.097$), PL vertical ($p = 0.022$, $\omega^2 = 0.072$), PL anteroposterior ($p = 0.013$, $\omega^2 = 0.086$), PL mediolateral ($p = 0.007$, $\omega^2 = 0.105$), and Power met ($p = 0.021$, $\omega^2 = 0.082$). This confirms that biological maturation accounted for a significant portion of the variance within the physical load data.

Post-hoc pairwise comparisons tracked a distinct, inverse behavioral trend, demonstrating that the elite team accumulated significantly fewer accumulated body loads than the sub-elite and, in most dimensions, amateur teams during match play (see Table 6). For total Player Load, the sub-elite cohort demonstrated large differences compared to the elite team (d = 1.049, $p = 0.013$), whereas non-significant moderate differences were identified between amateur and elite teams (d = 0.777, $p = 0.061$) and trivial differences were found between amateur and sub-elite players (d = 0.272, $p = 0.705$).

Table 6. Descriptive statistics (mean +/− sd) for impacts and load metrics for amateur, sub-elite and elite teams, with univariate ANCOVAs *p*-values, and effect size ω^2.

Variable	Amateur	Sub-Elite	Elite	*p*-Value	ω^2
Impacts					
Total Impacts	15,580 ± 3531	15,420 ± 4853	13,190 ± 3685	0.21	-
High Impacts Cnt	29.30 ± 18.53	27.59 ± 15.25	21.38 ± 17.18	0.256	-
Load					
Player Load (au)	101.6 ^{ab} ± 21.61	103.8 ^{b} ± 28.85	87.05 ^{a} ± 23.96	0.012 *	0.115
PL horizontal (au)	68.27 ^{a} ± 14.41	66.83 ^{a} ± 17.67	57.58 ^{b} ± 15.79	0.015 *	0.108
PL vertical (au)	63.82 ^{ab} ± 13.96	66.43 ^{b} ± 19.19	55.64 ^{a} ± 15.53	0.021	0.1
PL anteroposterior (au)	44.78 ^{a} ± 9.25	43.30 ^{a} ± 10.85	37.38 ^{b} ± 9.97	0.012 *	0.115
PL mediolateral (au)	41.52 ^{ab} ± 9.19	41.06 ^{b} ± 11.65	35.58 ^{a} ± 10.13	0.025 *	0.091
Power met	18,810 ± 2777	19,460 ± 4077	19,050 ± 2608	0.447	-
Power met avg	7.83 ± 0.74	8.046 ± 0.83	8.32 ± 0.80	0.218	-
HMLD	716.20 ± 271.20	871.10 ± 410.80	951.20 ± 232.30	0.073	-
Energy Expenditure	523.60 ± 128.00	587.50 ± 201.50	558.80 ± 111.40	0.193	-

Note: *p*-value refers to the *p*-value of the univariate ANCOVA test for the main effect of Team Level. $\omega 2$ represents the effect size; only reported for significant effects ($p < 0.05$). * $p < 0.05$. - value means not reported/calculated by JASP for non-significant results. Means that do not share a common superscript letter (a, b) are significantly different from one another according to Tukey's HSD post-hoc test.

4. Discussion

The aim of this study was to compare the locomotor demands of three teams competing at different levels within the same youth age category (U13). Both multivariate and univariate analyses revealed a consistent pattern: differences between competitive levels (amateur, sub-elite, and elite) did not manifest uniformly across all locomotor variables. Instead, group distinctions emerged predominantly in metrics that capture high-intensity actions and performance-oriented movement behaviors. These findings

underscore the multifactorial nature of physical performance in youth football and emphasize the need to differentiate between global load indicators and intensity-specific measures when attempting to characterize competitive level.

Across all teams, variables representing total running volume (e.g., TD, ED, and low-to-moderate speed running up to ~18 km·h^{-1}) showed no significant differences. Previous research comparing team level within the same age group is not recent and is limited and inconsistent. Waldron et al. [44] reported that elite players covered greater TD (115.7 ± 6.6 vs. 105.4 ± 7.7 m/min) than sub-elite players, while Randers et al. [45] found notable differences between competitive levels (2088 ± 299 vs. 1764 ± 307 m). These discrepancies likely stem from methodological variability—different match formats, durations, and contextual constraints, as well as limitations of earlier GPS technology, which used lower sampling rates and lower speed thresholds, probably underestimating high-speed activity. In our study, contextual factors such as match structure, team formation, pitch dimensions, and game duration were standardized, which may contribute to the similar total distances observed across competitive levels [46,47]. However, a clear differentiation emerges when examining higher velocity thresholds. Waldron et al. defined high-speed running as >19.1 km·h^{-1} and reported no differences between elite and sub-elite players (3.4 ± 0.7 vs. 3.2 ± 1.4 m/min in a 75-min match with four periods). By contrast, Randers et al., using >20 km·h^{-1}, found substantial differences between levels (69 ± 45 vs. 17 ± 22 m in 20-min 11v11 and 8v8 games). Although these studies offer contrasting results, the variability in speed thresholds, match duration, and recovery structure makes direct comparisons with our data difficult. Despite this, our findings reveal a clear inflection point above 18 km·h^{-1}. Initial differences appear between amateur and elite players and become more pronounced beyond 24 km·h^{-1}, where elite players cover substantially greater distances than sub-elite and, in particular, amateur players. Locomotor demands above 30 km·h^{-1}, virtually absent in amateur players and only sporadic in sub-elite players, are consistently present only in the elite group matches, providing a strong discriminator of competitive level. Notably, since the covariate PHV demonstrated no significant association with these high-speed metrics, this competitive edge cannot be attributed to differences in biological maturation, suggesting instead a genuine superiority in physical performance capacity and match-related physical efficiency among elite players. Overall, these results indicate that high-speed and very-high-speed running distances are key distinguishing characteristics of more proficient youth players, suggesting that competitive differentiation in U13–U14 football is primarily reflected in the ability to reach and repeatedly utilize maximal and submaximal velocities.

A similar pattern emerged for acceleration–deceleration demands. Previous research in youth soccer has shown that the frequency and intensity of acceleration and deceleration actions vary considerably across match conditions [48]. Comparisons between small-sided games and peak match periods further indicate that only specific game formats reproduce acceleration loads comparable to competition, emphasizing the importance of movement intensity rather than total event counts [49]. Similarly, professional-level evidence suggests that cumulative acceleration–deceleration load may remain relatively stable across the training–match cycle, whereas its distribution across intensity zones varies considerably [50]. Consequently, aggregated measures such as average acceleration or deceleration may be less sensitive to differences occurring at the upper end of the intensity spectrum.

Our findings are consistent with this perspective. Amateur and sub-elite players accumulated more acceleration and deceleration events, whereas elite players achieved higher maximal acceleration and deceleration values and covered greater distances within high-intensity acceleration zones. Thus, differences among competitive groups were more evident in the magnitude of high-intensity actions than in their overall frequency. These

variables may therefore provide more specific information for distinguishing locomotor profiles across competitive levels. While the theoretical literature suggests that biological development contributes substantially to high-intensity performance, our ANCOVA models effectively neutralized the influence of biological maturation (PHV). Despite this control, the elite players still demonstrated a clear superiority in maximal values and high-intensity acceleration zones, confirming that competitive level exerts an independent effect on these neuromuscular metrics. Sprint-related outcomes showed a comparable pattern. Relative sprint measures and average speed did not differ significantly among competitive levels, whereas absolute sprint distance, absolute HSR, and maximal speed showed clearer between-group differences, particularly between elite and amateur players. These findings suggest that absolute high-speed measures may be more sensitive than relative measures for characterizing the competitive groups examined. The use of absolute versus relative thresholds remains an important methodological consideration. Absolute thresholds facilitate between-player comparisons but may underestimate the relative demands experienced by younger or less mature players who are unable to reach equivalent absolute running velocities [51]. Furthermore, acceleration capacity partly depends on the velocity at which an action begins, as accelerations initiated at moderate or high speeds have less opportunity to reach predefined absolute thresholds [52]. Previous studies have also demonstrated age- and maturation-related increases in sprint capacity, with older and more mature players generally achieving higher running velocities and sprint distances [53,54]. This is particularly relevant because the elite group in this study was chronologically older and biologically more mature than the amateur group.

Importantly, the greater absolute sprint and HSR values observed in elite players reflect a clear, independent effect of competitive standard that transcends the underlying influence of biological growth. Although our ANCOVA models detected that the covariate PHV was significantly associated with these absolute volumetric metrics ($p < 0.05$), the analytical control effectively neutralized this maturational bias. Crucially, even after accounting for the variance explained by biological development, the independent effect of competitive level remained highly significant and substantial for absolute sprint ($p < 0.001$, $\omega^2 = 0.197$) and HSR distances ($p = 0.003$, $\omega^2 = 0.16$). Previous research suggests that absolute thresholds may be particularly useful for between-player or between-group comparisons, whereas individualized relative thresholds may provide complementary information when monitoring high-velocity exposure within players [55]. Thus, absolute and relative approaches should be regarded as complementary strategies whose usefulness depends on the purpose of the assessment. No significant differences were observed in total or high-intensity impact metrics across competitive levels. Impact exposure may be influenced by individual and match-related characteristics, including age, body mass, playing time, and the frequency of physical contacts and duels [56–59]. The absence of significant differences therefore suggests that impact variables were less sensitive than high-intensity locomotor measures for distinguishing the three groups. Conversely, this statistical behavior shifts fundamentally when examining structural body loads, where biological maturation acts as a critical confounding factor rather than a parallel contributor.

In contrast to the preliminary univariate findings, the ANCOVA models revealed that aggregated load variables, specifically total Player Load and its multi-planar vector components (horizontal, vertical, anteroposterior, and mediolateral), did significantly differ among groups. This underscores a critical methodological insight: biological development accounted for a substantial portion of the variance, meaning that a standard ANOVA completely obscured the true effect of competitive standard under a blanket of maturational noise. Interestingly, the post-hoc trends revealed that elite players accumulated significantly lower body loads than their sub-elite and amateur counterparts.

These measures combine several movement components and may therefore be less sensitive to specific high-intensity actions. Reche-Soto et al. [60] reported differences in PL, MP, and HMLD between starters and substitutes in semi-professional soccer, but also strong relationships among these measures, suggesting that they predominantly represent overall match load. Acceleration-derived metrics may also obscure differences in the direction and intensity of individual actions [61], while contextual factors may further influence their expression [62]. Consistently, Casamichana et al. [63] found no differences in HMLD between teams from the Spanish first and second divisions. Overall, global load metrics, when properly adjusted for biological maturation, showed capacity to differentiate the competitive groups examined in youth soccer, suggesting that elite players navigate match demands with optimized structural and mechanical efficiency. In contrast, differences were more consistently observed at the upper end of the locomotor intensity spectrum, particularly for absolute high-speed running, maximal speed, and high-intensity acceleration–deceleration measures. These findings support examining specific high-intensity variables alongside aggregated load measures when profiling match demands in youth soccer, while accounting for developmental characteristics and match context.

4.1. Strengths and Limitations

A key strength of this study is the standardized setting used to compare players across competitive levels. All participants belonged to the same club and shared a common developmental structure, while relevant match characteristics (including playing format, pitch dimensions, match duration, and team formation), were consistent across groups. This approach reduced several sources of between-team variability that commonly affect comparisons involving different clubs. Moreover, locomotor data were obtained during official matches using GPS technology, providing an ecologically valid representation of the physical demands experienced in competitive youth soccer.

Nevertheless, several limitations should be considered when interpreting the findings. First, the sample was relatively small and derived from a single club, which may limit the generalizability of the results to other teams, competitive structures, or age categories. Although the within-club design increased methodological consistency, future multicenter studies involving larger samples are needed to determine whether similar locomotor patterns are observed across different youth soccer environments.

Second, although the distinct competitive cohorts inherently diverged in chronological age and somatic maturity, the implementation of ANCOVA models with PHV as a continuous covariate allowed us to statistically isolate the unique impact of competitive standard. While crude physical outputs in youth sports are typically confounded by biological growth, this analytical control ensured that our findings reflect genuine performance differences rather than mere maturational advantages. Nevertheless, future research using maturity-matched samples or longitudinal designs remains recommended to fully clarify the respective contributions of competitive standard and biological development over time.

Third, although several match conditions were standardized, other contextual factors that may influence locomotor performance were not included in the analyses. These include playing position, opponent quality, match location, score status, and ball possession. The number of matches analyzed also differed slightly among teams. Accordingly, some of the observed variability may have been influenced by match-specific or positional factors that could not be accounted for in this study.

Fourth, the complete physical fitness testing battery was not available for the elite group. Due to logistical constraints and more limited access to these players, physical tests such as the Yo-Yo IR1, 10-m sprint, 505 change-of-direction, and repeated-sprint ability

tests could not be administered. Consequently, these variables could not be compared across all three competitive levels or comprehensively related to match-derived locomotor performance. This limits the possibility of comparing how components of physical fitness can be integrated into the interpretation of the between-group differences.

Finally, the cross-sectional design does not allow for causal or predictive conclusions. Although several high-intensity locomotor variables differed among the competitive groups, these findings do not demonstrate that such variables predict future selection, progression, or sporting success. Therefore, absolute high-speed running, maximal speed, and acceleration–deceleration measures should be considered complementary variables for player profiling rather than validated markers of talent identification. Larger longitudinal studies involving multiple clubs, maturity-matched players, and positional and match-context information are warranted to determine the developmental and predictive relevance of these locomotor characteristics.

4.2. Practical Applications

Absolute high-speed running, maximal speed, and high-intensity acceleration–deceleration measures and multi-planar aggregated load metrics may provide useful complementary information when profiling match demands across competitive levels. In contrast, total movement volume and aggregated load metrics appear less sensitive to these differences. Practitioners should interpret absolute and relative thresholds according to the purpose of monitoring and the developmental characteristics of each player. Absolute measures may facilitate between-group comparisons, whereas relative thresholds can provide a more individualized representation of locomotor demands. Importantly, these variables should be interpreted alongside chronological age, biological maturity, and other dimensions of performance rather than used independently for talent identification or selection.

5. Conclusions

This study showed that absolute high-speed running, sprint distance, maximal speed, high-intensity acceleration–deceleration variables and multi-planar aggregated load metrics differed among amateur, sub-elite, and elite youth soccer teams, whereas total running volume and several aggregated load metrics showed limited discriminatory capacity. While the naturally occurring groups differed in chronological age and biological maturation, the application of ANCOVA models with PHV as a continuous covariate successfully isolated the independent effect of competitive standard across these high-intensity and multi-planar mechanical metrics. Accordingly, high-intensity locomotor variables and adjusted body load profiles may provide useful complementary information for profiling youth soccer players, but their potential role in talent identification requires confirmation through larger, maturity-matched and longitudinal investigations.

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Institutional Review Board Statement: The study was conducted in accordance with the ethical principles of the Helsinki Declaration for human research and was approved by the Research Ethics Committee of the University of Granada (no. 3882/CEIH/2023). After obtaining approval, the authors invited all the coaches of the teams and families to a meeting, during which we presented the

objectives of the project and asked them to sign an informed consent form. Parents, teams officials, and coaches were informed that they could revoke the participation agreement.

Informed Consent Statement: Informed consent was obtained from all subjects involved in the study.

Data Availability Statement: The original contributions presented in the study are included in the article, further inquiries can be directed to the corresponding author.

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